

Statistical Concepts Used in Lab 4

Brief revision guide — Medical Insurance Costs

Based on the statistical requirements in the Lab-4 assignment: descriptive statistics, hypothesis testing, OLS regression and diagnostic checks. ■filecite■turn3file0■L30-L44■

1. Descriptive Statistics

Concept	Brief explanation
Mean	Average: sum of observations divided by number of observations.
Median	Middle value after sorting; less affected by extreme values than the mean.
Standard deviation	Measures spread around the mean. Larger SD means more variability.
IQR	Interquartile Range = $Q3 - Q1$; spread of the middle 50% of data.
Skewness	Measures asymmetry. Positive = longer right tail; negative = longer left tail.
Kurtosis	Describes tail heaviness/peakedness relative to a normal distribution.

2. Distribution & Relationship Plots

Concept	Brief explanation
Histogram	Shows the distribution/frequency of a numerical variable.
KDE	Kernel Density Estimate: smooth estimate of a numerical distribution.
Scatter plot	Shows relationship/trend between two numerical variables.
Box plot	Shows median, quartiles, spread and possible outliers; useful for comparing groups.
Correlation	Measures strength and direction of a linear relationship; ranges from -1 to $+1$.

3. Hypothesis Testing Basics

Concept	Brief explanation
H_0 (null hypothesis)	Default claim, commonly 'no difference' or 'no association'.
H_1 (alternative)	Claim that a difference or association exists.
$\alpha = 0.05$	Significance level required in this assignment.
p-value	Probability of a result at least this extreme assuming H_0 is true.
Decision	$p < 0.05 \rightarrow$ Reject H_0 . $p \geq 0.05 \rightarrow$ Fail to Reject H_0 .
Important	Failing to reject H_0 does not prove H_0 is true.

4. Shapiro–Wilk Normality Test

Concept	Brief explanation
Purpose	Checks whether data are consistent with a normal distribution.
H_0	The data follow a normal distribution.
H_1	The data do not follow a normal distribution.
Decision	$p < 0.05 \rightarrow$ reject normality; $p \geq 0.05 \rightarrow$ insufficient evidence against normality.
Use here	The assignment uses this before choosing the two-group test. ■filecite■turn3file0■L33-L38■

5. Levene's Test

Concept	Brief explanation
Purpose	Checks whether groups have equal variances.
H_0	Group variances are equal.
H_A	At least one variance differs.
Decision	$p < 0.05 \rightarrow$ evidence of unequal variances; $p \geq 0.05 \rightarrow$ insufficient evidence of unequal variances.

6. Two-Sample t-Test

Concept	Brief explanation
Purpose	Compares means of two independent groups when the outcome is reasonably normal.
H_0	$\mu_1 = \mu_2$.
H_A	$\mu_1 \neq \mu_2$ for a two-sided test.
Idea	Difference between sample means is compared with its standard error.
Formula	$t = (\bar{x}_1 - \bar{x}_2) / SE(\bar{x}_1 - \bar{x}_2)$.

7. Mann–Whitney U Test

Concept	Brief explanation
Purpose	Non-parametric alternative for comparing two independent groups when normality is not appropriate.
Idea	Uses ranks rather than requiring normally distributed raw values.
H_0	The two groups have the same distribution (usual formulation).
Decision	Small p-value $\rightarrow$ evidence that the groups differ.
Assignment rule	If the two groups are not normal, use Mann–Whitney U. ■filecite■turn3file0■L35-L38■

8. One-Way ANOVA

Concept	Brief explanation
Purpose	Tests whether means of 3 or more independent groups are equal.
H_0	$\mu_1 = \mu_2 = \dots = \mu_k$.
H_A	At least one group mean differs.
F statistic	Between-group variability divided by within-group variability.
Important	A significant ANOVA says at least one mean differs; it does not identify the pair. A post-hoc test such as Tukey can do that.
Assignment example	Compare average charges across the four regions. ■filecite■turn3file0■L39-L42■

9. Multiple Linear Regression (OLS)

Concept	Brief explanation
Purpose	Predict/model a numerical response using several predictors.
Model	$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \epsilon$
Y	Response/dependent variable; here, medical charges.
β_0	Intercept.
β_i	Expected change in Y for a one-unit increase in X_i , holding other predictors constant.

Concept	Brief explanation
ε	Random error/residual.
OLS	Ordinary Least Squares estimates coefficients by minimizing the sum of squared residuals. The assignment requires statsmodels.api.OLS. ■filecite■turn3file1■L69-L70■

10. Dummy Variables

Concept	Brief explanation
Why	Categorical predictors such as sex, smoker and region must be represented numerically for OLS.
Reference category	One category is treated as the baseline.
Coefficient	Shows the difference from the reference category, holding other predictors constant.
Drop-first	Leaving one category out avoids perfect multicollinearity.

11. Coefficients, p-values & Confidence Intervals

Concept	Brief explanation
Coefficient	Estimated effect of a predictor while controlling for other predictors.
Coefficient p-value	Tests $H_0: \beta_j = 0$.
95% CI	Range of plausible values for a population coefficient.
Significance	For a two-sided 5% test, a 95% CI excluding 0 corresponds to statistical significance.

12. R^2 and Adjusted R^2

Concept	Brief explanation
R^2	Proportion of variation in the response explained by the fitted predictors.
Formula	$R^2 = 1 - SSE/SST$.
Adjusted R^2	Like R^2 but penalizes unnecessary predictors; useful when comparing models with different numbers of predictors.
Caution	High R^2 does not automatically mean a model is appropriate or causal.

13. Residuals & Fitted Values

Concept	Brief explanation
Residual	Observed value minus predicted value: $e_i = y_i - \hat{y}_i$.
Fitted value	Predicted value produced by the regression model.
Residual analysis	Used to assess whether regression assumptions are reasonably satisfied.

14. Linearity & Homoscedasticity

Concept	Brief explanation
Linearity	The systematic relationship should be adequately represented by the linear model.
Homoscedasticity	Residual variance should be approximately constant across fitted values.
Residuals vs fitted	Random cloud around zero is desirable. Curves suggest non-linearity; a funnel pattern suggests changing variance.
Assignment	Residuals vs fitted plot is required. ■filecite■turn3file1■L71-L74■

15. Q-Q Plot & Residual Normality

Concept	Brief explanation
Q-Q plot	Compares ordered residuals with theoretical normal quantiles.
Good pattern	Points roughly follow the reference line.
Departure	Strong curvature or tail deviations suggest non-normality.
Why	Residual normality is one of the diagnostic checks required by the assignment.

16. Jarque–Bera / Omnibus

Concept	Brief explanation
Purpose	Formal tests used to assess residual normality.
Jarque–Bera	Uses skewness and kurtosis to assess departure from normality.
Decision	$p < 0.05 \rightarrow$ evidence against normal residuality; $p \geq 0.05 \rightarrow$ insufficient evidence against it.
Assignment	Q-Q plot plus Jarque–Bera or Omnibus is required. ■filecite■turn3file1■L71-L74■

17. Multicollinearity & VIF

Concept	Brief explanation
Multicollinearity	Predictors contain overlapping information and are strongly linearly related.
VIF	Variance Inflation Factor measures inflation of coefficient variance due to predictor relationships.
Formula	$VIF = 1 / (1 - R^2)$.
Rule of thumb	$VIF \approx 1$ is ideal; >5 is often treated as a warning and >10 as serious, though these are guidelines.
Assignment	VIF is required for continuous predictors. ■filecite■turn3file1■L71-L74■

18. Confidence vs Prediction Interval

Concept	Brief explanation
Confidence interval	Uncertainty interval for the mean response at specified predictor values.
Prediction interval	Interval for a new individual observation; wider because it includes individual random variation.
Assignment	The live prediction tab requires a 95% confidence/prediction interval. ■filecite■turn3file2■L105-L110■

19. Quick Test-Selection Flow

Concept	Brief explanation
Two groups + normal	Two-sample t-test.
Two groups + non-normal	Mann–Whitney U.
Three or more groups + numerical outcome	One-way ANOVA.
Two categorical variables	Chi-square test is the other option specified in the assignment.
Regression	Use multiple OLS for a numerical response and then check residuals and VIF.

Most Important Formulas

$$\text{Mean} = \Sigma x / n$$

$$\text{IQR} = Q3 - Q1$$

$$\text{Residual} = y - \hat{y}$$

$$R^2 = 1 - \text{SSE}/\text{SST}$$

$$\text{Regression: } Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$$

$$\text{VIF}_j = 1 / (1 - R_j^2)$$

Exam tip: Always state H_0 , H_a , α , test statistic/p-value, decision, and conclusion in a hypothesis-test answer.